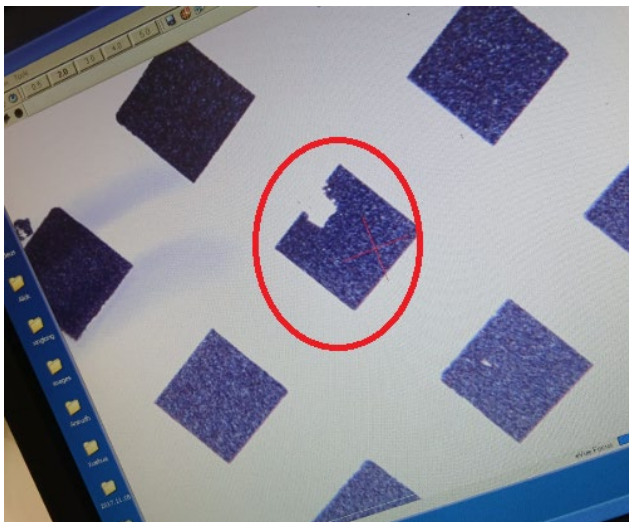


Term 7: Nanoelectronics and Technology

Course project Part 2:

I would like the project discussion to address the following points:

1. **Dielectric degradation and breakdown in 10 nm SiO_2**
Perform I–V and I–t measurements, analyze the results, extract key reliability parameters, and explain the underlying physical mechanisms.
2. **Analysis of the ALD-grown Al_2O_3 sample**
 - (a) Using the provided measurement files, estimate the oxide thickness, assuming a breakdown electric field of 8 MV/cm.
 - (b) If the intended ALD thickness was 10 nm, discuss the possible reasons for any discrepancy between the target value and the experimentally extracted thickness.
3. **Observation of damaged metal electrode pads**
In some samples, the metal pads appear as shown in the figures after electrical characterization. Discuss the possible causes of this behaviour.



4. **Thickness deviation in e-beam-evaporated Ta_2O_5**
While targeting 10 nm thickness using e-beam evaporation, the resulting physical thickness was 25 nm. Explain the possible reasons for this deviation.
5. Supplementary simulations or comparisons with SiO_2 data would further strengthen the analysis."